

Christopher Musco

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Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. Candidate, Computer Science – Theory of Computation

2013 – present

Advisors: Jonathan Kelner, Martin Rinard

S.M. Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Yale University

New Haven, CT

B.S. Applied Mathematics, B.S. Computer Science

2008 – 2012

Appl. Math Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

Comp. Sci. Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

Research Interests

I am interested in dimensionality reduction and its application to randomized numerical linear algebra, sketching and streaming algorithms, and machine learning. I also work on problems related to algorithmic graph theory and collaborate with MIT's Center for Resilient Software on quantitative approaches to computer security and approximate computation.

Academic Positions

Massachusetts Institute of Technology

Cambridge, MA

Research Assistant

2013 – present

Yahoo! Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Teaching and Mentorship

Research Science Institute/MIT

2014

Summer Research Advisor

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Designed a project on concurrent data structures under the direction of Professor Martin Rinard.
- Chris' paper, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Yale University

Teaching Assistant, Introduction to Computer Science

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Teaching Assistant, Introduction to Programming

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

Undergraduate Grader, Introductory Calculus

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Honors and Awards

National Science Foundation 2014
Graduate Research Fellowship

Yale University 2012
Cum Laude, Distinction in Both Majors

Publications

Conference Papers.....

[Stronger and Faster Approximate Singular Value Decomposition via the Block Lanczos Method](#). Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 papers).

[Dimensionality Reduction for K-Means Clustering and Low Rank Approximation](#). Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

[Principled Sampling for Anomaly Detection](#). Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

[Uniform Sampling for Matrix Approximation](#). Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

[Single Pass Spectral Sparsification in Dynamic Streams](#). Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. Invited to SICOMP Special Issue.

Talks

Large Scale Column Subset Selection
Yahoo Labs, Science Week Poster Session July 2015

Dimensionality Reduction for K-Means Clustering
IBM T.J. Watson, Research Seminar Aug. 2015
Symposium on Theory of Computing (STOC) June 2015
MIT, Theory Lunch June 2014

Principled Sampling for Anomaly Detection
MIT, Defense Advanced Research Projects Agency Site Visit Mar. 2015
Network and Distributed System Security Symposium (NDSS) Feb. 2015

Uniform Sampling for Matrix Approximation
Yahoo Labs, Scalable Machine Learning Research Seminar June 2015

Single Pass Spectral Sparsification in Dynamic Streams
MIT, Annual Sublinear Algorithms Day (Poster Session) Apr. 2015
Harvard University, Theory Seminar Nov. 2014
Foundations of Computer Science (FOCS) Oct. 2014

Service

External Reviewer.....
Symposium on Discrete Algorithms (SODA) 2016
Neural Information Processing Systems (NIPS) 2015

European Symposia on Algorithms (ESA)	2015
Foundations of Computer Science (FOCS)	2015
International Colloquium on Automata, Languages and Programming (ICALP)	2015
Network and Distributed System Security Symposium (NDSS)	2015

Professional Experience

Redfin **Seattle, WA**
Software Engineer, Data Team *2012 – 2014*

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code base.

Elysium Digital **Cambridge, MA**
Summer Consultant *Summer 2011*

Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus **New Haven, CT**
Software Developer *2010 – 2011*

Built tools for political campaign management and social fundraising at early-stage startup.

Moody's Investors Service **New York, NY**
Intern, Asset Finance Group *Summer 2009*

Established and monitored bond ratings on asset securitizations.

U.S. Department of Agriculture **East Greenwich, RI**
Intern, Natural Resources Conservation Service *Summer 2007*

Collected and analyzed data for state conservation and research projects.